

Improving Intersection Traffic Flow Using Deep Q-Learning Algorithms

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Abstract—Efficient control strategies play a vital role in mitigating traffic congestion in urban areas. This study introduces an adaptive traffic signal control approach using Deep Q-Learning (DQN) family techniques, such as, the Double DQN variant. The conventional fixed-time control strategies, whose fixed signal timings often cause inefficiency, are usually not suitable, whereas the models in this paper can use real-time traffic data to achieve the regulated phases of signals in an optimal order. DQN improves decision-making by learning optimal signal times, while Double DQN further improves stability and accuracy by rectifying the Q-value overestimation issue by means of decoupled action selection and evaluation. It learns to improve signal timings based on changing traffic patterns. Experiments conducted using the SUMO simulator demonstrate that the proposed framework significantly reduces cumulative delay by 75% and queue length by 78%. The models are considerably better than fixed-timing schemes and traditional Q-Learning methods, and Double DQN shows better adaptability and stability under oscillating traffic conditions.

Index Terms—Intelligent Signal Control, Traffic Signal Optimization, Deep Reinforcement Learning, Double Q-Networks, Urban Mobility Simulation

I. INTRODUCTION

Rising vehicle ownership and fast urbanization are aggravating traffic congestion in metropolitan areas all around the world. Roads getting more crowded affects longer travel times, more fuel use, and growing pollution levels. Conventional traffic control systems[4] mostly depend on fixed timing cycles[21], thus the dynamic and complicated flow of vehicles is usually too much for them. Particularly during peak hours. Commuters are firmly impacted, and inefficiency happen due to these systems not being flexible enough to react to traffic conditions in real time. Thus, the demand for intelligent and sophisticated traffic management solutions has never been more as cities try to solve the increasing complexity and limit of urban traffic networks.

Factors such as time of day, and real time incidents e.g., accidents, and road work, cause effects that are persistent and cumulative [13]. Traffic flow changes during different times

of the day in high-density vehicle areas such as intersections within commercial or residential areas. Keeping effective signal timing becomes important to keep bottlenecks from occurring and to smoother flow of traffic. Imbalance of vehicle distribution due to dynamic change of locality-based traffic pattern usually leads to traffic jam in some areas combined with wastage in other areas. Through timely setting of signal timing to respond to the current conditions, many of these problems can be accommodated or mitigation applied.

Traditional signal control systems, based on pre-determined signal plans, have considerable challenges in addressing such increasingly dynamic traffic conditions. Machine Learning (ML) is capable of handling these limitations by using real-time data for traffic signal control optimization [16]. ML models predict and optimize signal times by prioritizing key considerations such as vehicle flow and speed. It allows traffic systems to respond to present conditions, reducing wasted time and congestion for improved traffic management. Nonetheless, ML models usually require manual feature engineering and are prone to failure to find non-linear relationships in traffic patterns, therefore they often fail when facing the large-scale, high-dimensional traffic systems.

These problems can be addressed in a more sophisticated way by using DL. By using deep neural networks, capable of automatically learn hierarchically-level features from large-scale of real-time traffic information, DL can effectively deal with such complex high-dimensional data. This aids in making accurate determination on traffic signal control. In near real-time dynamically adjusting signal timings are assigned by DL approaches in highly densified and dynamic urban environments where traditional ML models can not perform well, which helps traffic flow smooth, congestion and disruptions are mitigated and congestion are reduced [23].

RL takes the concept further with the more powerful characteristics compared to DL in adaptive traffic signal control. The capability of learning optimal strategies is achieved through interacting with a network environment, and adapting signal

timings according to the real-time feedback using reinforcement learning (RL). However, the conventional RL methods often rely on predetermining time durations to switch phases, possibly limiting their agility during heavy-traffic periods. Moreover, advanced methods, Like deep reinforcement learning (DRL), using deep neural networks for reinforcement learning, are capable to tackle large state spaces and continuously make the best real-time decisions. These DRL-based techniques enhance traffic signal control's reactivity and efficiency, therefore addressing the limits of conventional reinforcement learning [27].

II. LITERATURE REVIEW

The real-time, stochastic changes in urban traffic flows are often not captured. Supported by traditional conventional traffic signal control [19], that mostly operate on deterministic time-based cycles or rules-based approach sets. These cause overcrowding and long waits failure by systems to respond correctly to surprise such as accidents, peak traffic, or construction projects [4], [21]. Allowing agents to find the strongest and most helpful types of trial-and-error strategies to interact with their environment (although Reinforcement Learning (RL) remains a more flexible framework. The agent iteratively updates its policy to minimize delays and enhance processing capacity while monitoring the status of the traffic, and selecting action (e.g. to alter the phase of the signal), being paid according to the traffic on the network generated [30].

Reinforcement Learning and meta-heuristics such as Particle Swarm Optimization (PSO) and Genetic Algorithms (GA) have also been hybridized for enhanced adaptively in traffic control. Using these approaches, the RL agent can escape local optimums and efficiently converge to global solutions when solving high-dimensional, non-trivial optimization problems [7]. When RL combines meta-heuristic [8], we can get a very strong hybrid model with the capability of adjusting signal timings dynamically according to the real-time traffic information, especially in upgrading scale and congestion control.

The large and unsolvable non-linear state spaces, which are the typical representation of these systems, are challenging for classical RL approaches to deal with. This is targeted by Deep Reinforcement Learning (DRL) (and in particular Deep Q-Learning (DQN)) [18], wherein deep neural networks are employed to approximate Q-values across the state-action space [24]. DRL based traffic signal control DRL methods are suitable for adaptive signal control in practical environments because they are able to learn the complex traffic pattern from the raw data directly without the heavy handcrafted feature engineering [15], [9]. These models have shown potential to significantly reduce queue lengths and wait time by adjusting signal timings according to real-time values of traffic.

Since DQN shares the same network between learning and acting, it is biased toward overestimating the value of an action. The objective of Double Deep Q-Learning (Double DQN) is to decouple these two functions (use one network for the evaluation of the action and one to choose them) [20].

This update has the advantage of both stabilising learning and producing more accurate value estimates. Double DQN is known to achieve higher convergence rates and better traffic signal control performance (i.e., reducing the sum of the cumulative delays and the queue lengths) in traffic circumstances, particularly in the case of the oscillating traffic condition.

While single-agent DRL techniques work well at isolated intersections, they are unable to take inter-intersection dependencies into consideration. By assigning multiple agents, each in charge of an intersection, the ability to exchange information and cooperatively optimize their strategies, Multi-Agent Reinforcement Learning (MARL) expands the advantages of DRL to coordinated traffic networks [31], [11]. Global traffic flow optimization is made possible by this cooperative learning, especially during erratic occurrences like emergencies or high-volume surges [12]. By coordinating decision-making across intersections, MARL systems have demonstrated promise in lowering network-wide congestion [26], [1].

III. PROPOSED METHODOLOGY

This section presents the design and implementation of the proposed adaptive traffic signal control system using Deep Q-Learning (DQN) and Double Deep Q-Learning (Double DQN). The approach includes network architecture, learning algorithms, reward formulation, and computational optimizations, evaluated using a simulated SUMO environment.

A. System Overview and Environment Setup

This figure visually represents the simulation setup used in SUMO. The key components state space, queue length, and traffic signal phases are labeled as they form the input and control variables used by the DQN and Double DQN agents. The state vector is dynamically generated from these observations and fed into the learning model for decision-making.

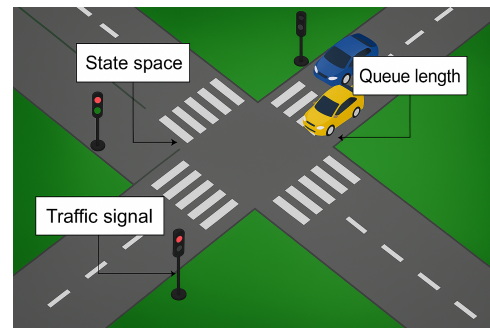


Fig. 1. Intersection layout highlighting state space, queue length, and traffic signals for RL-based control.

B. Deep Q-Learning (DQN) Framework

DQN employs a neural network to estimate the state-action utility, aiming to converge toward the optimal policy over time. The update of Q-values during training is driven by the Bellman equation:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right] \quad (1)$$

where s is the current state, a is the action taken, r is the immediate reward, s' is the next state, α is the learning rate, and γ is the discount factor. The goal is to minimize the temporal difference (TD) error between the predicted Q-value and the target Q-value.

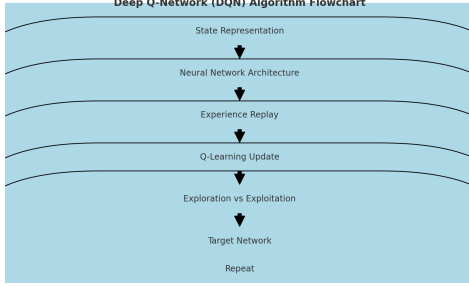


Fig. 2. Deep Q-Network (DQN) architecture illustrating state-action mapping.

The DQN model uses a replay memory to store experience tuples and trains the Q-network using mini-batches sampled randomly, enhancing learning stability.

C. Double Deep Q-Learning (Double DQN) Framework

Double Deep Q-Learning (Double DQN) addresses the overestimation bias present in traditional DQN by decoupling action selection from action evaluation. The learning process modifies the target value computation while keeping the rest of the DQN pipeline intact.

The target Q-value in Double DQN is estimated as follows:

$$y = r + \gamma Q(s', \arg \max_{a'} Q(s', a'; \theta); \theta^-) \quad (2)$$

where the action is selected using the online network parameters θ , but evaluated using the target network parameters θ^- . This separation reduces the risk of overestimating Q-values and results in more stable training.

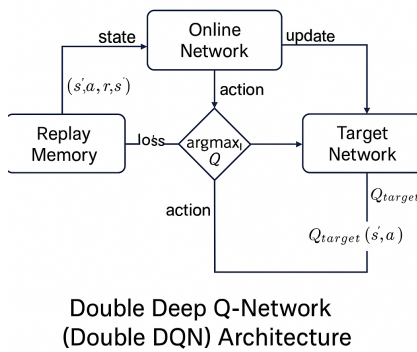


Fig. 3. Double DQN Architecture showing decoupling of action selection and evaluation.

This architecture enables more stable convergence compared to standard DQN.

D. Comparison of DQN and Double DQN Target Updates

The traditional DQN derives the target value by taking the maximum estimated Q-value of the target network (as shown in Equation 1). This method often overestimates the value because the action selection and evaluation share the same network.

To address this limitation, Double DQN separates these steps via two different networks. This action is determined using the online network (as shown in Equation 2) and its value is assessed with the target network. This diminishes the overestimation bias and enhances training stability.

E. Adaptive Signal Control with Real-Time Data

The agent receives traffic features such as vehicle count, waiting time, and queue length, forming a state vector. Based on this state, actions such as signal phase extension or switching are selected. The reward function is designed to minimize congestion:

$$R = -(w_1 \cdot QL + w_2 \cdot WT + w_3 \cdot CD) \quad (3)$$

where QL is queue length, WT is waiting time, CD is cumulative delay, and w_1, w_2, w_3 are weights.

F. Managing Large State Spaces

Urban traffic environments exhibit high-dimensional and non-linear state spaces due to the complex interactions between multiple traffic parameters such as vehicle count, queue length, and waiting times across several lanes. Traditional tabular Q-learning methods become infeasible in such settings due to the curse of dimensionality and poor generalization.

To effectively handle this complexity, Deep Q-Learning (DQN) and Double DQN leverage deep neural networks as function approximators for the action-value function $Q(s, a)$. These networks are capable of learning abstract representations of traffic patterns, enabling the agent to generalize to unseen states without exhaustive enumeration.

Two critical strategies are employed to stabilize learning in large state spaces:

- **Experience Replay:** The agent stores transition tuples (s, a, r, s') in a memory buffer and randomly samples them during training to reduce correlation between samples and enhance data efficiency.
- **Target Network Updates:** A distinct target network is employed to generate stable Q-value targets, with infrequent updates helping to reduce learning instability and prevent divergence.

By combining these techniques, the agent effectively manages high-dimensional input spaces and learns robust policies that adapt to diverse and dynamic traffic scenarios [25].

G. State and Action Space

The state representation includes: vehicle count per lane, queue length, and average waiting time, captured at 1-second intervals. Actions correspond to selecting one of the available phase configurations at a given time step. The state space is continuous and processed via a neural network, and the action space is discretized to fixed phase selection options to reduce dimensionality.

H. Optimizing Computational Efficiency

Real-time traffic signal control requires learning algorithms that balance accuracy with computational efficiency. To meet the constraints of real-world deployment, especially in embedded or edge-based traffic control systems, several optimizations are employed in the proposed framework:

- **Experience Replay:** By storing and reusing past transitions in a memory buffer, redundant online computations are avoided. This improves sample efficiency and reduces the frequency of costly environment interactions.
- **Target Network Updates:** Instead of updating Q-values with a frequently shifting target, a separate target network is updated at fixed intervals. This reduces the frequency of backward passes and stabilizes gradient updates, minimizing unnecessary computations.
- **Double DQN Decoupling:** By decoupling action selection and evaluation, Double DQN avoids Q-value overestimation. As a result, the learning process becomes more stable, requiring fewer episodes to reach convergence and reducing computational overhead.

These strategies collectively reduce training time, enhance convergence speed, and ensure that the model can be deployed in time-sensitive applications [22]. The framework is thus suitable for deployment on real-time traffic control systems, even with limited computational resources.

IV. RESULTS

The performance of the proposed DQN and Double DQN-based traffic signal control systems was evaluated with the help of the Simulation of Urban Mobility (SUMO) environment. Several metrics were used to assess the efficiency and learning behavior of the models, including cumulative delay, average queue length, and cumulative reward across training episodes.

A. Cumulative Delay Reduction

Cumulative delay indicates the total waiting amount of vehicles in intersections. Fig. 4 demonstrates the reduction in cumulative delay through training episodes of DQN and Double DQN. Initially, the high delay was due to random exploration. But with the progress of training both models achieved a steady decrease of delay and Double DQN, in particular, reduces delay up to 75%, outperforming DQN by approximately 5%.

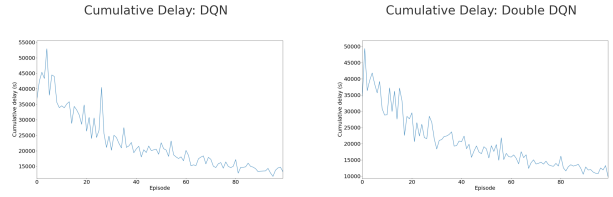


Fig. 4. Cumulative Delay vs. Episodes for DQN and Double DQN.

B. Queue Length Optimization

The reduction in average queue length is a direct measure of improved traffic flow. As shown in Fig. 5, the Double DQN model consistently outperformed the DQN model, stabilizing at a queue length of 4.16 vehicles per lane compared to 4.69 for DQN, indicating a 11.3% improvement.

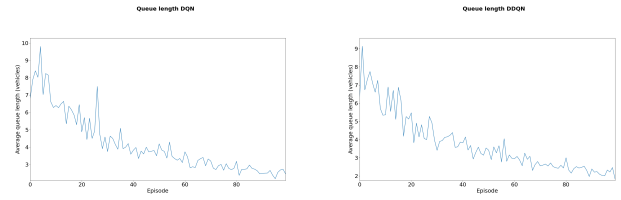


Fig. 5. Queue Length vs. Time Steps for DQN and Double DQN.

C. Cumulative Reward Convergence

Cumulative reward reflects the model's ability to learn and apply optimal actions over time. As illustrated in Fig. 6, both DQN and Double DQN showed steady reward improvement during training. However, Double DQN achieved faster convergence and higher reward values, owing to its decoupled action evaluation mechanism, which reduces Q-value overestimation and enhances policy stability.

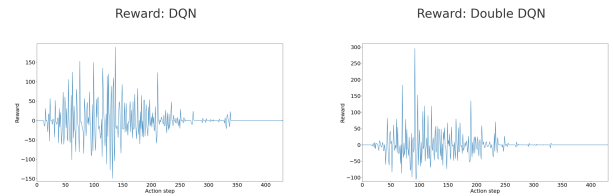


Fig. 6. Cumulative Reward vs. Episodes for DQN and Double DQN.

D. Model Comparison

Table I summarizes the quantitative performance metrics for all models, including fixed-time and conventional Q-Learning, highlighting the significant improvements achieved by DQN and Double DQN models.

TABLE I
COMPARISON OF TRAFFIC CONTROL MODELS

Metric	Fixed-Time	Q-Learning	DQN	Double DQN
Convergence Speed	N/A	200 eps	100 eps	80 eps
Avg Queue Length	12.0	6.5	4.69	4.16
Cumulative Delay (s)	45000	20000	15000	10000
Max Queue Length	30.0	20.0	17.5	12.8
Avg Waiting Time (s)	45	25	15	10
Adaptability	None	Partial	Full	Full, Stable

V. LIMITATIONS

Although our Double DQN model has shown significant enhancement over the baseline methods, the current model is limited to a single intersection, and the external environment is just a controllable SUMO. Physical noise, communication delay and non-cooperative drivers are not considered in the models. Further, the performance of the model has not been evaluated at over-saturated (or at unpredictable) traffic conditions. For future work, the scalability of the approach to multi intersection networks and its real-time adaptation will be considered.

VI. CONCLUSION AND FUTURE WORK

An adaptive signal control model was developed based on Deep Q-Learning (DQN) and its Double DQN extension. Experimental results on SUMO-based simulation environment demonstrated significant improvement on flow efficiency of traffic. Between traditional DQN and fixed-time strategies, the Double DQN consistently benefited from a more stable learning process, shorter queue length, lower aggregate delay and faster convergence rate. The suggested system demonstrates its resilience and scalability for real-time urban traffic management by efficiently managing high-dimensional state spaces and mitigating Q-value overestimation bias.

Extending this model to multi-intersection scenarios using Multi- Agent Reinforcement Learning (MARL) can help to coordinate traffic flow over networks in next projects. Moreover, including Vehicle-to- Infrastructure (V2I) connection and real-time sensor data helps to increase system responsiveness. To make the system more inclusive and sustainable for smart city applications, other enhancements might involve adding more reward components like emissions, fuel consumption, and pedestrian flow [14].

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